

CIVL4340

Wind Engineering

Week 8: Bluff body aerodynamics

Overview

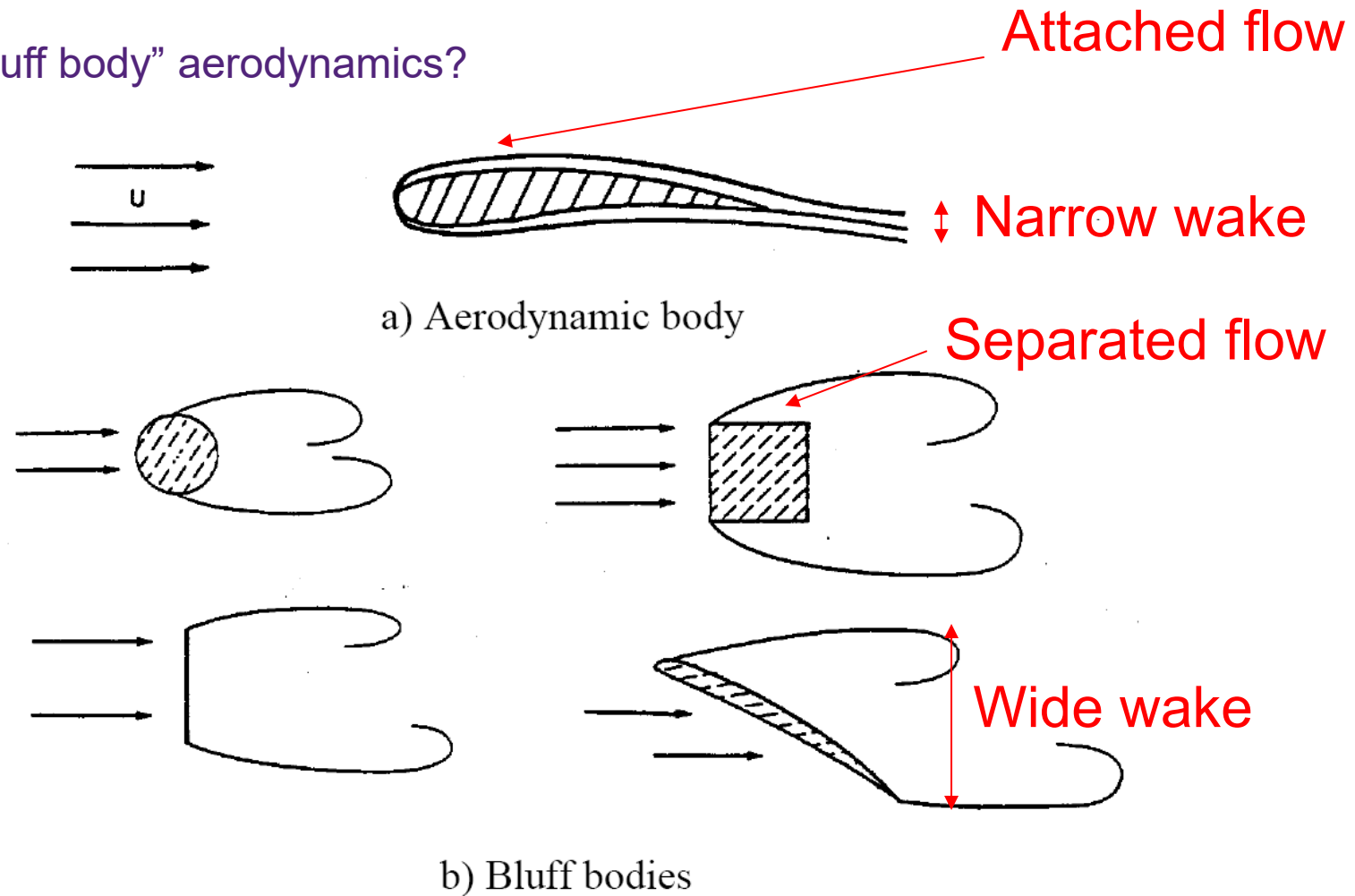
- Introduction to bluff bodies – what are they?
- Pressures and pressure coefficients.
- Integrated loads – drag and lift.
- Time-dependent (fluctuating) loads.
- How does AS/NZS1170.2 deal with bluff bodies?

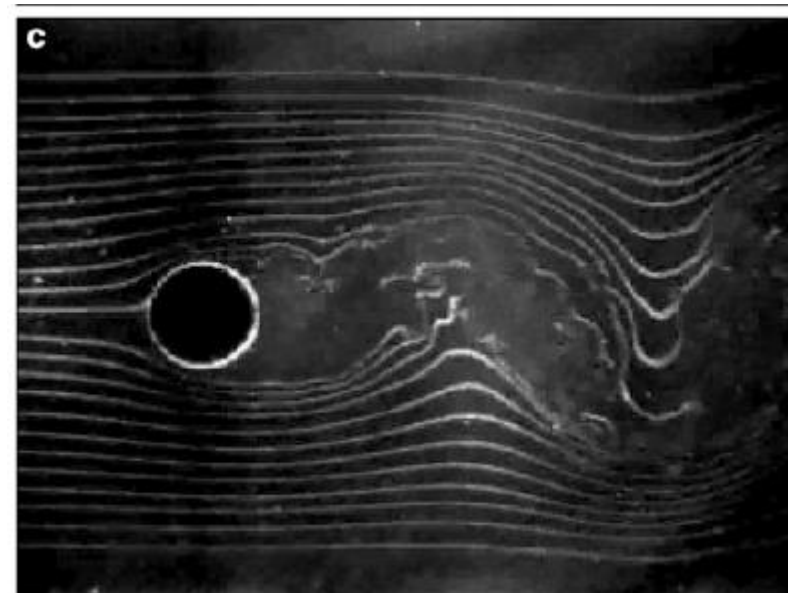
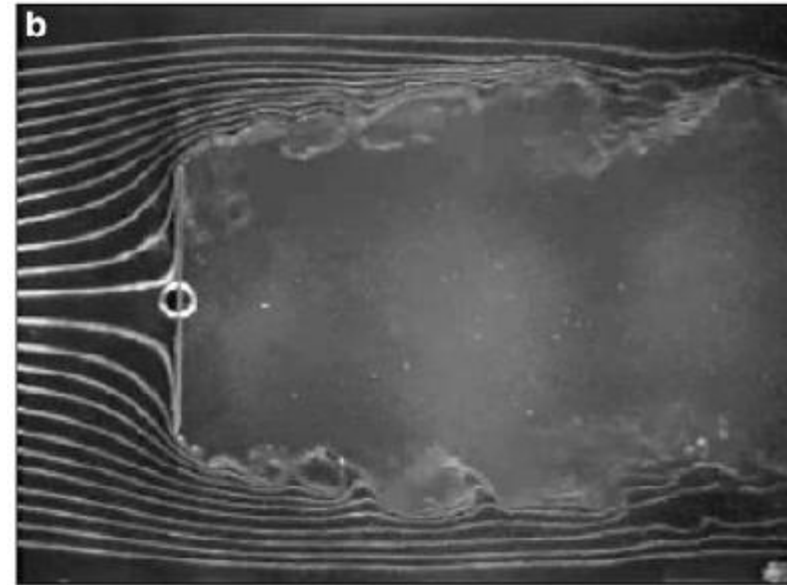
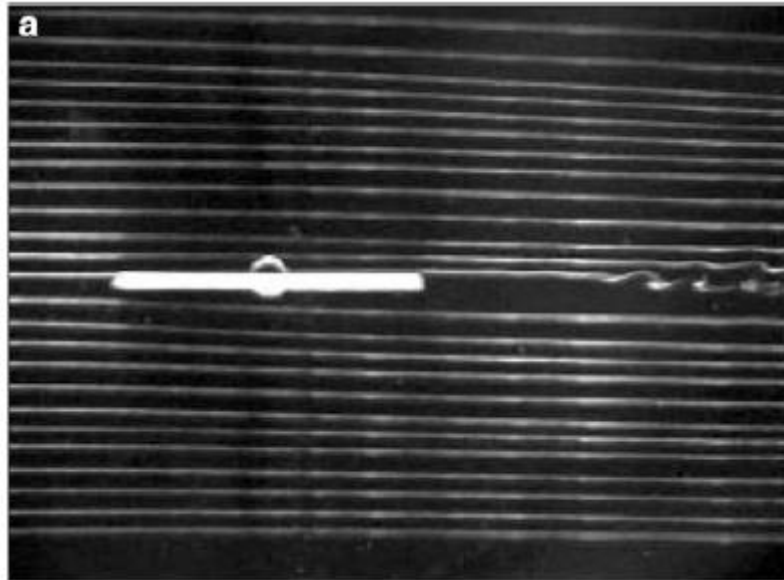
Reading

- Holmes textbook, Chapter 4.
- Simiu and Yao, Chapter 4.
- AS/NZS1170.2, Chapter 5.

Bluff body aerodynamics

What do we mean by “bluff body” aerodynamics?



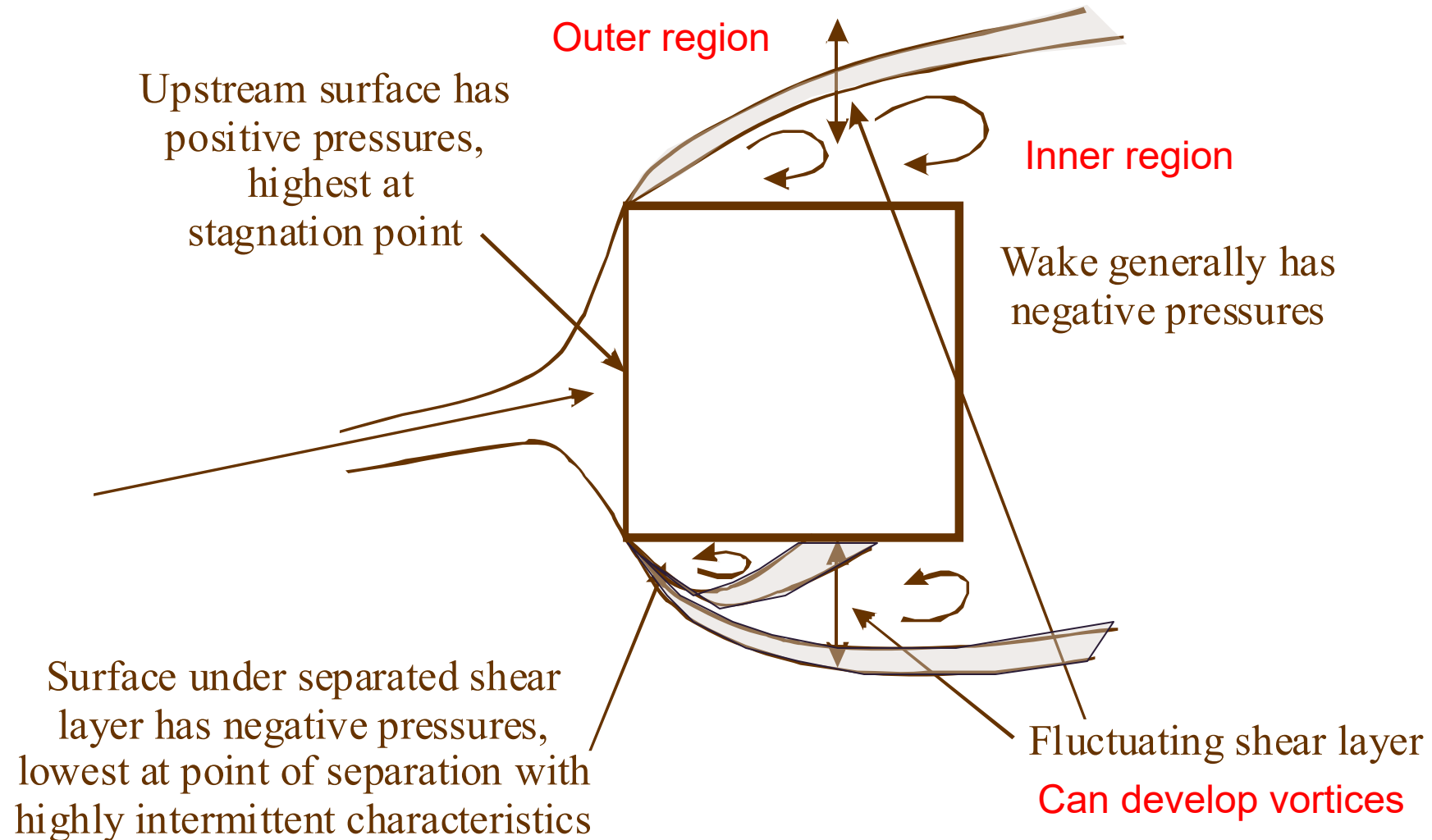


Shear layer



<https://www.youtube.com/watch?v=UEgk2Bgz16s>

Bluff body aerodynamics



Pressure coefficients

Outer region

- Inviscid (zero viscosity), irrotational (zero rotation)

$$z + p + \frac{1}{2} \rho_a \bar{U}^2 = \text{constant}$$

along a streamline

Freestream p, U

$$p_0 + \frac{1}{2} \rho_a \bar{U}_0^2 = p + \frac{1}{2} \rho_a \bar{U}^2$$

assuming the elevation is constant

Pressure form

$$C_p = \frac{p - p_0}{\frac{1}{2} \rho_a U_0^2} = \frac{\frac{1}{2} \rho_a (U_0^2 - U^2)}{\frac{1}{2} \rho_a U_0^2} = 1 - \left(\frac{U}{U_0} \right)^2$$

Velocity form

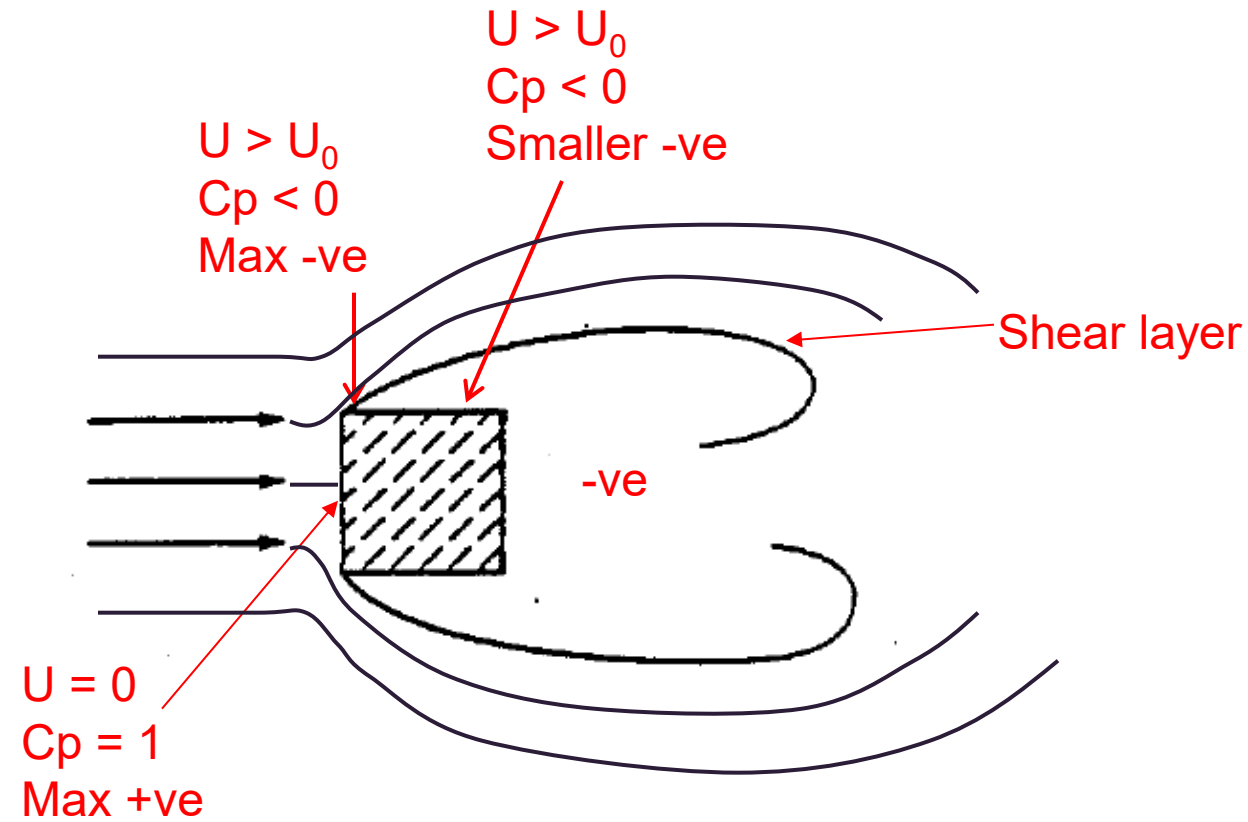
$$U = 0 \quad C_p = 1.0 \quad (\text{stagnation point}) ; U > U_0 \quad C_p < 0$$

Pressure coefficients

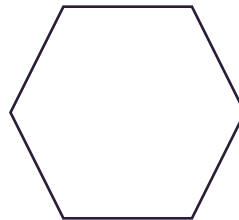
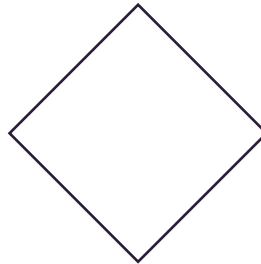
Inner region

- Not simple, dependent on body shape, turbulence, time, position....
- No analytical solutions.
- However, pressure is reasonably constant throughout the wake and can be estimated by using the relationship on the previous slide just outside the wake shear layer.

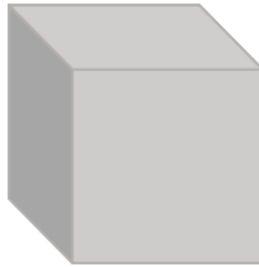
Pressure coefficients



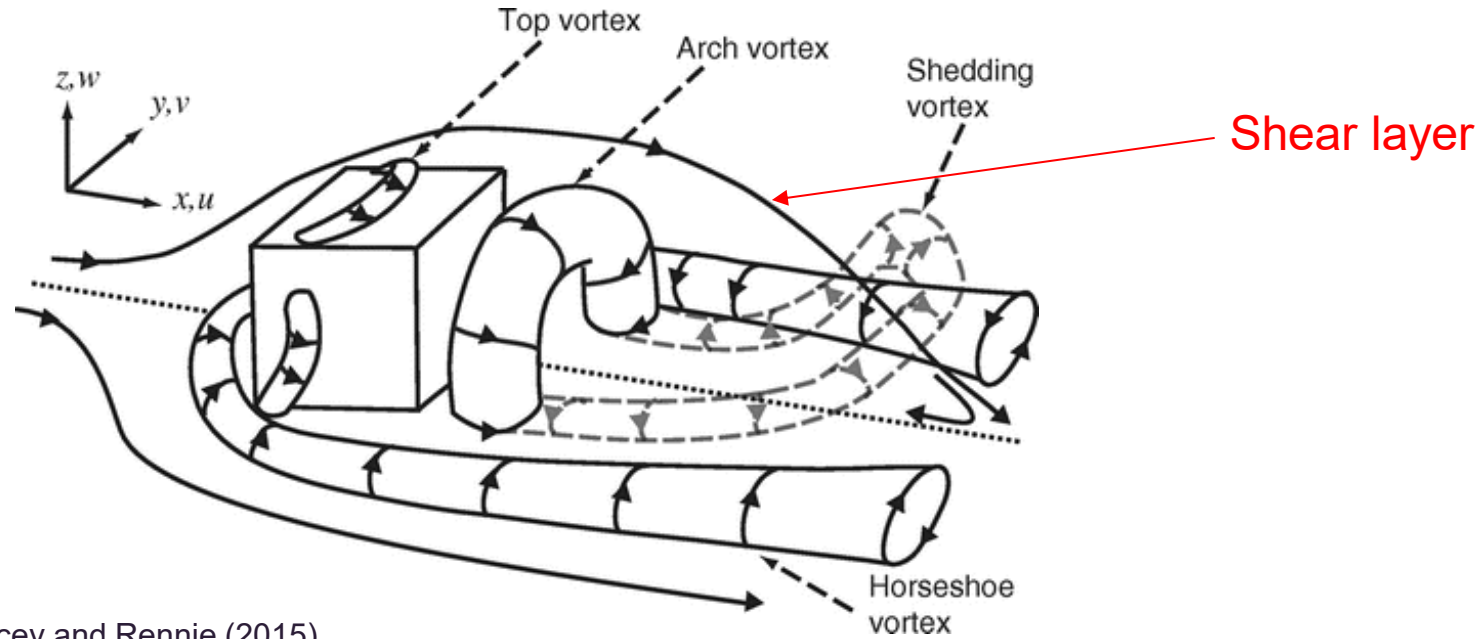
Pressure coefficients



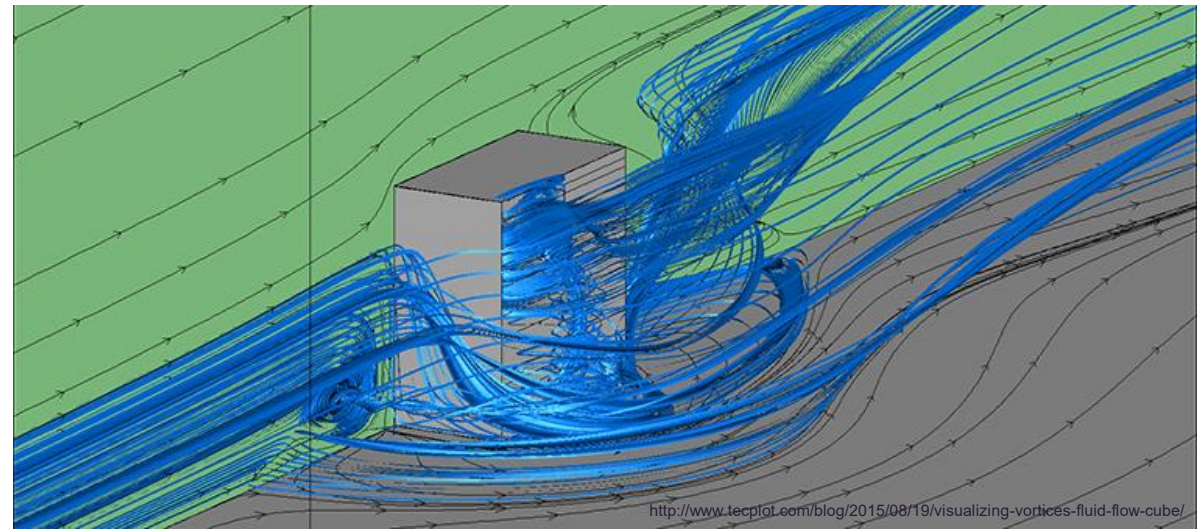
Pressure coefficients



Pressure coefficients

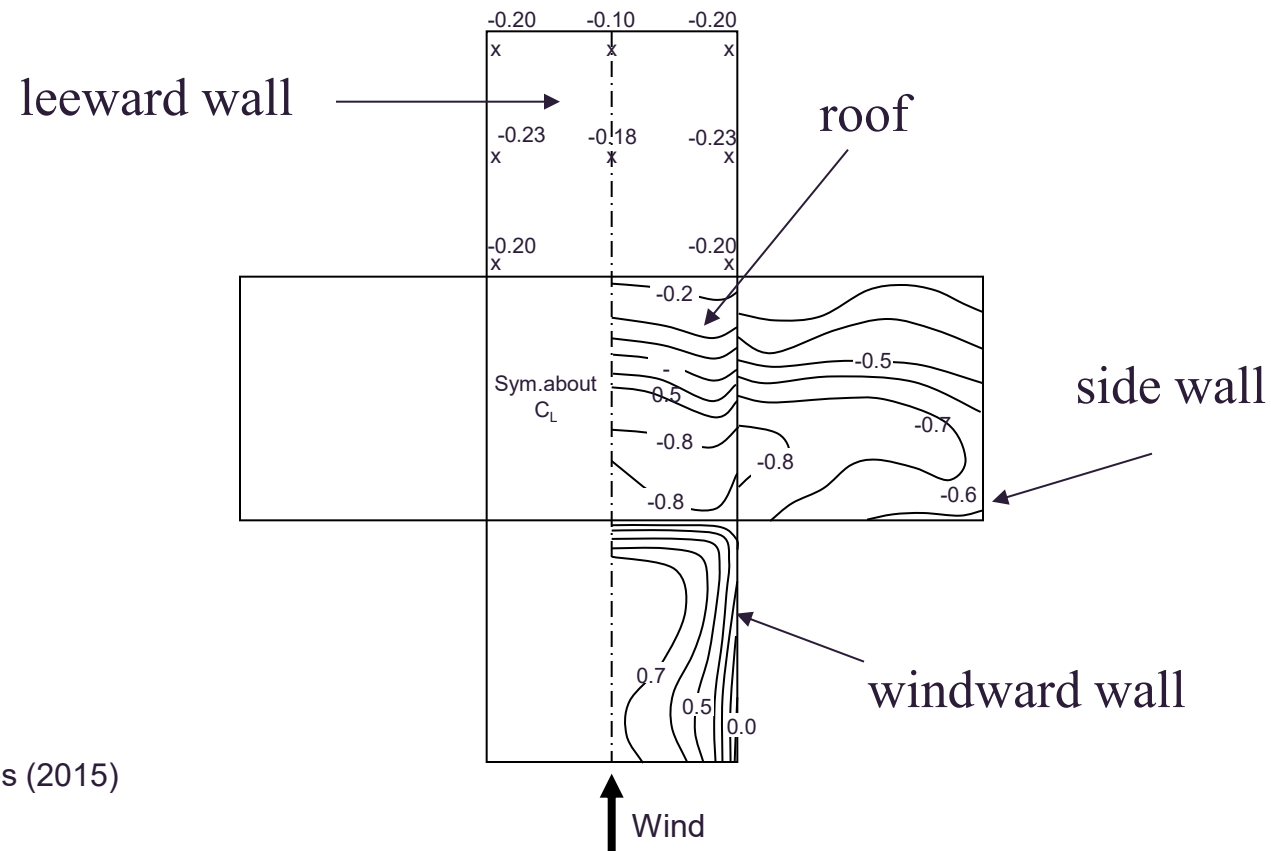


Lacey and Rennie (2015)



Pressure coefficients

3D boundary layer

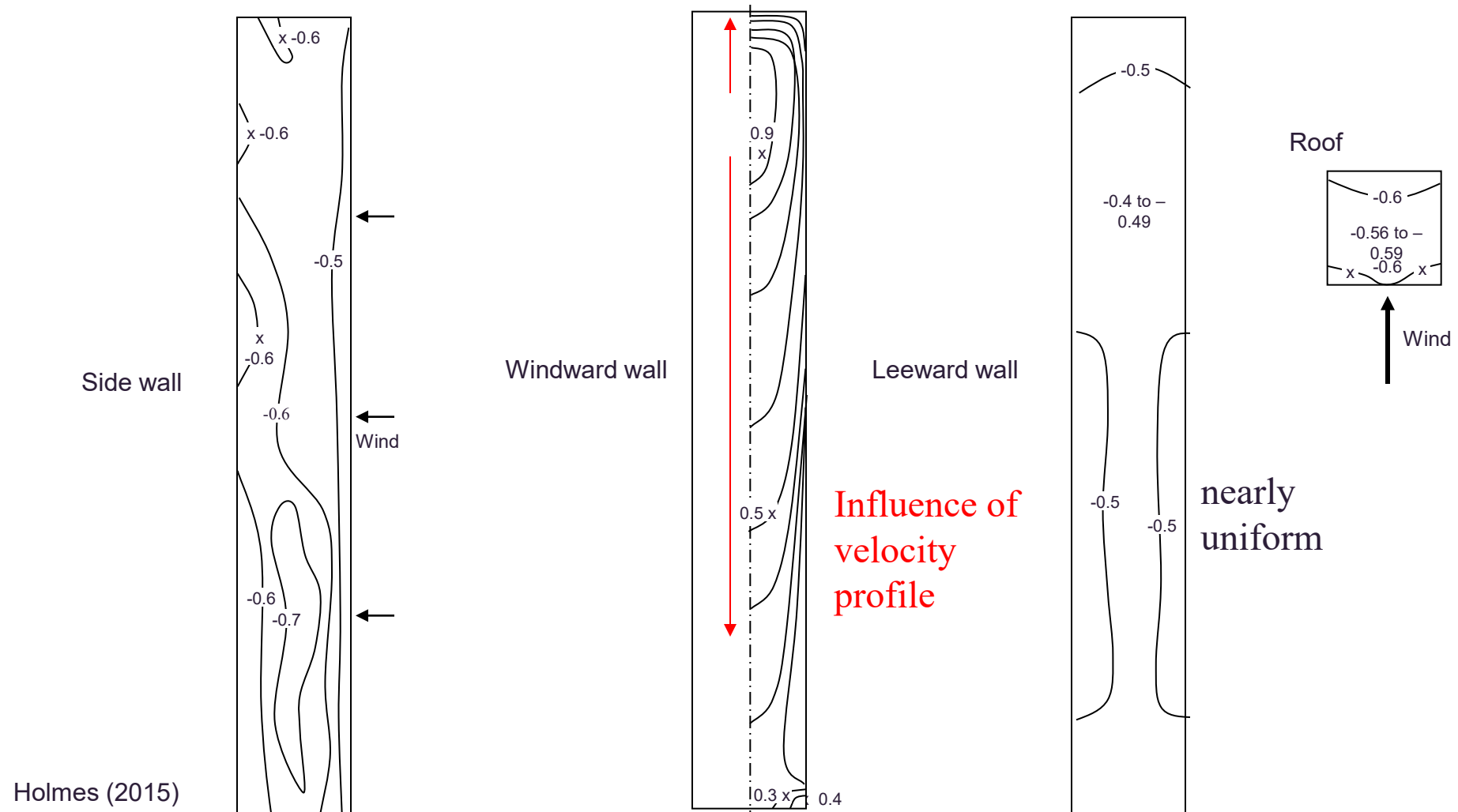


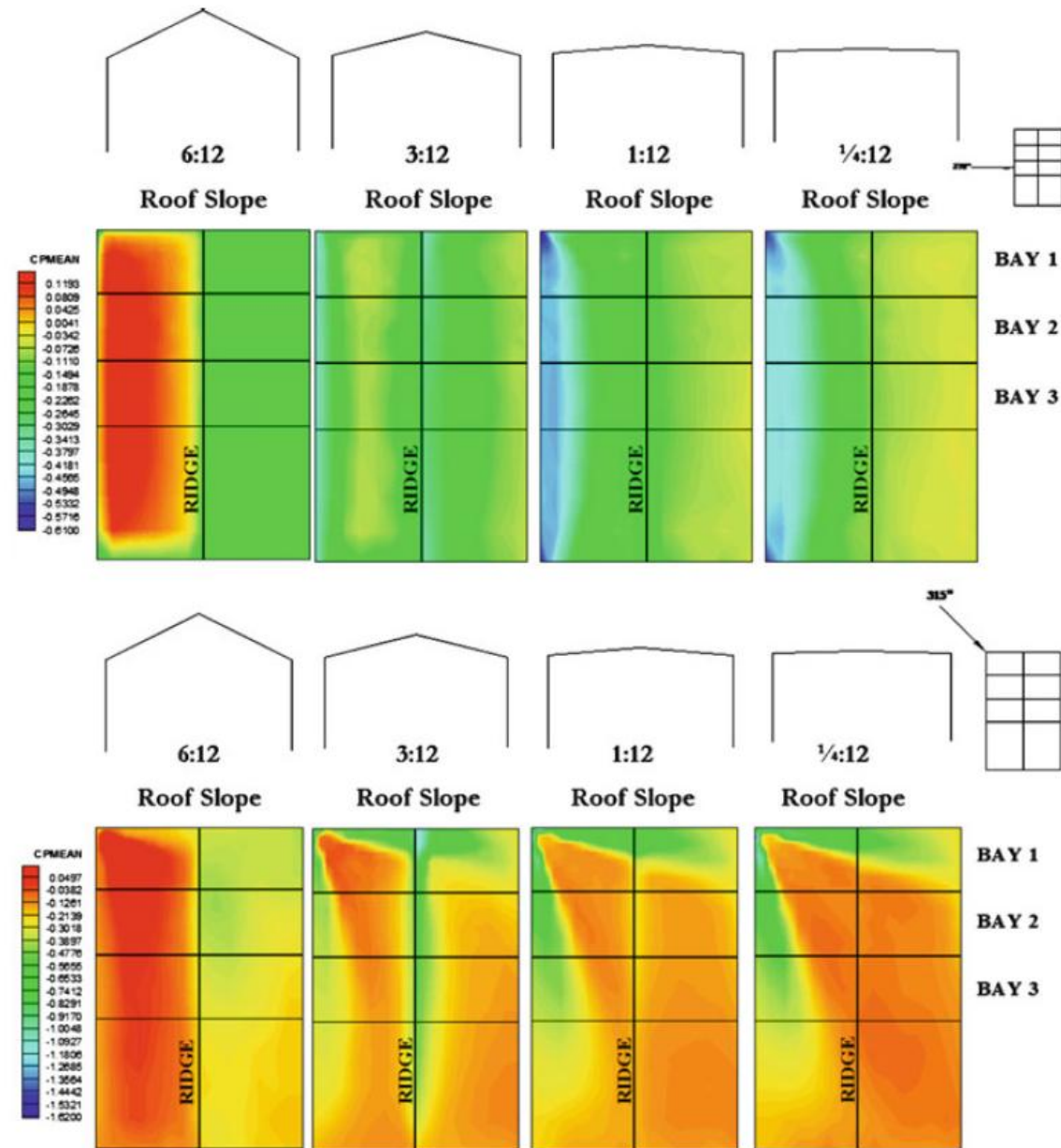
Holmes (2015)

Pressure coefficients



Pressure coefficients





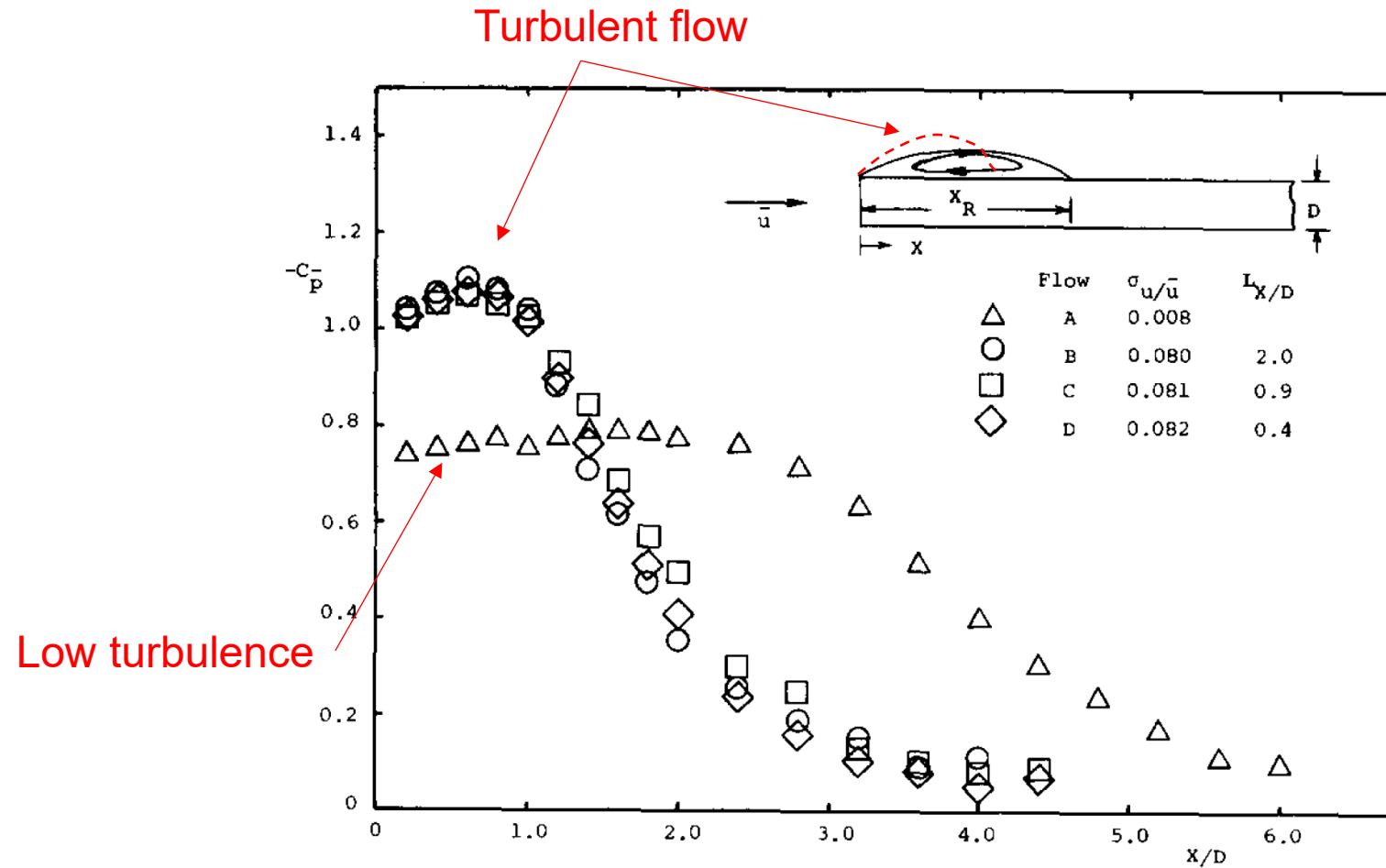
Tamura and Kareem (2013)

Fig. 7.3 Mean pressure coefficients on gable roofed buildings with different roof slopes for two different wind directions

Influence of Turbulence

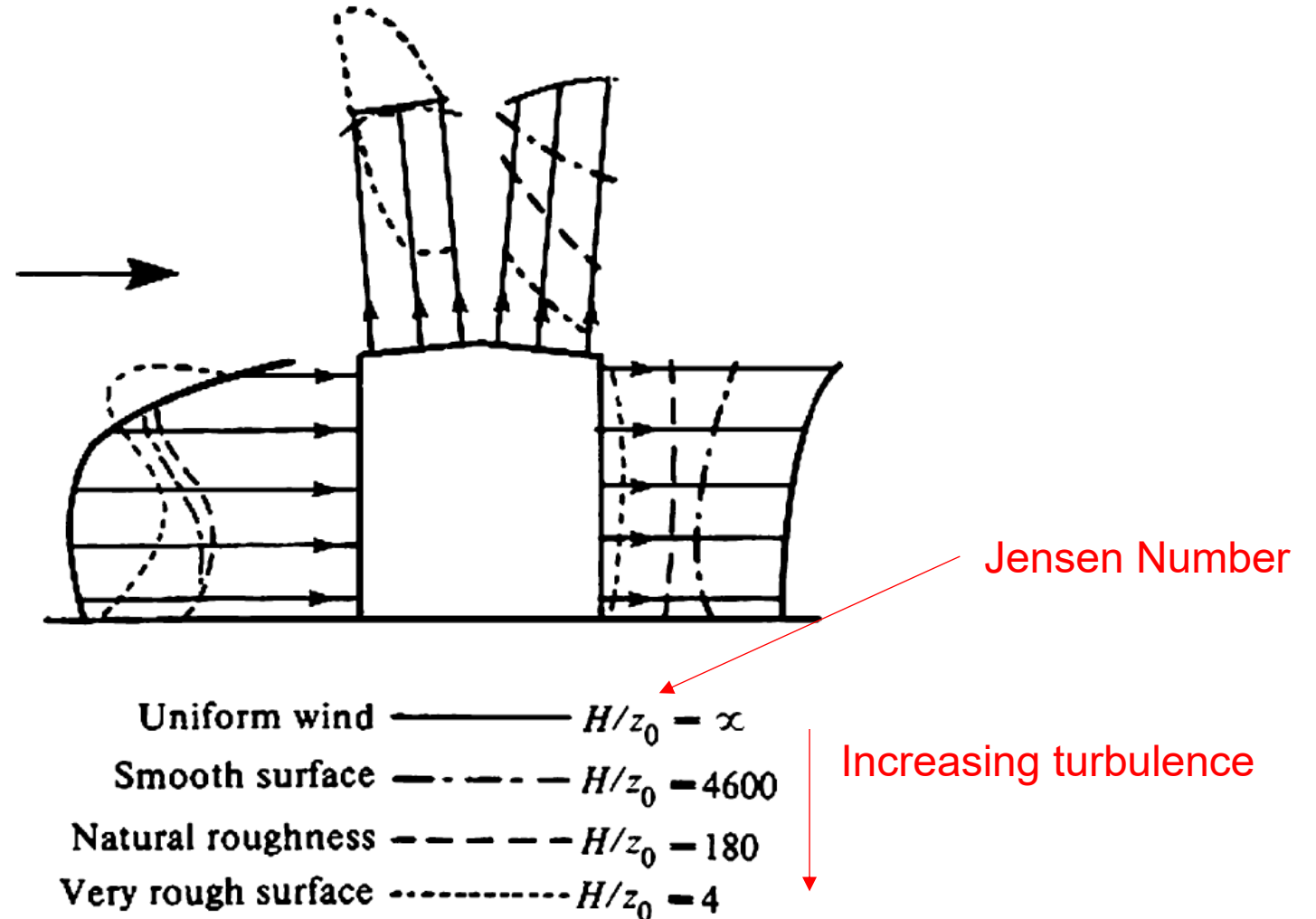
- Turbulence plays a critical role in defining the separated shear layer and the pressures within the wake.
- Therefore, bluff bodies experience different pressure loads in turbulent and non-turbulent winds.
- Turbulence increases wake fluid entrained into the shear layer, which reduces wake pressure (and loads on body faces in this region).
- Turbulence may also promote earlier reattachment of flow to bodies with long afterbodies.

Influence of turbulence



Influence of turbulence

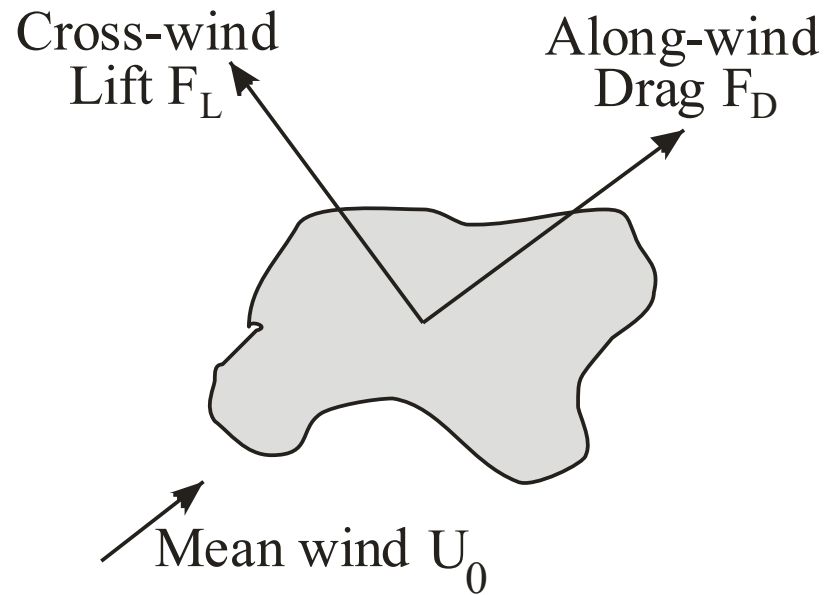
Fig. 3.17 Effect of surrounding ground roughness on the pressure distribution over a house, where H is height of house and z_0 is the roughness length (after Scruton 1981)



Forces

Integrating **pressure** over the structure give forces ($F=PA$)

Additional force due to **viscous shear** (generally much smaller)

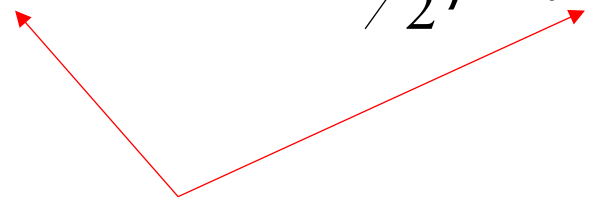


Force coefficients

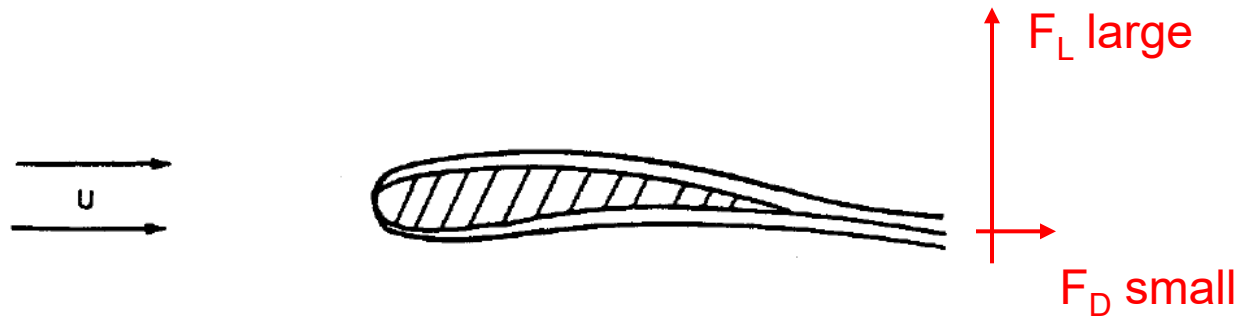
- As with pressures, forces presented as coefficients
- Describes the loading of a body not a location

$$C_D = \frac{F_{Drag}}{\frac{1}{2} \rho \bar{U}_0^2 A} \qquad C_L = \frac{F_{Lift}}{\frac{1}{2} \rho \bar{U}_0^2 A}$$

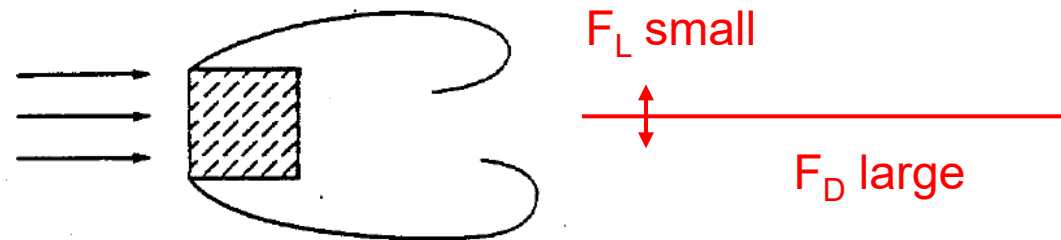
Area (not always the same)



(Mean) Force coefficients

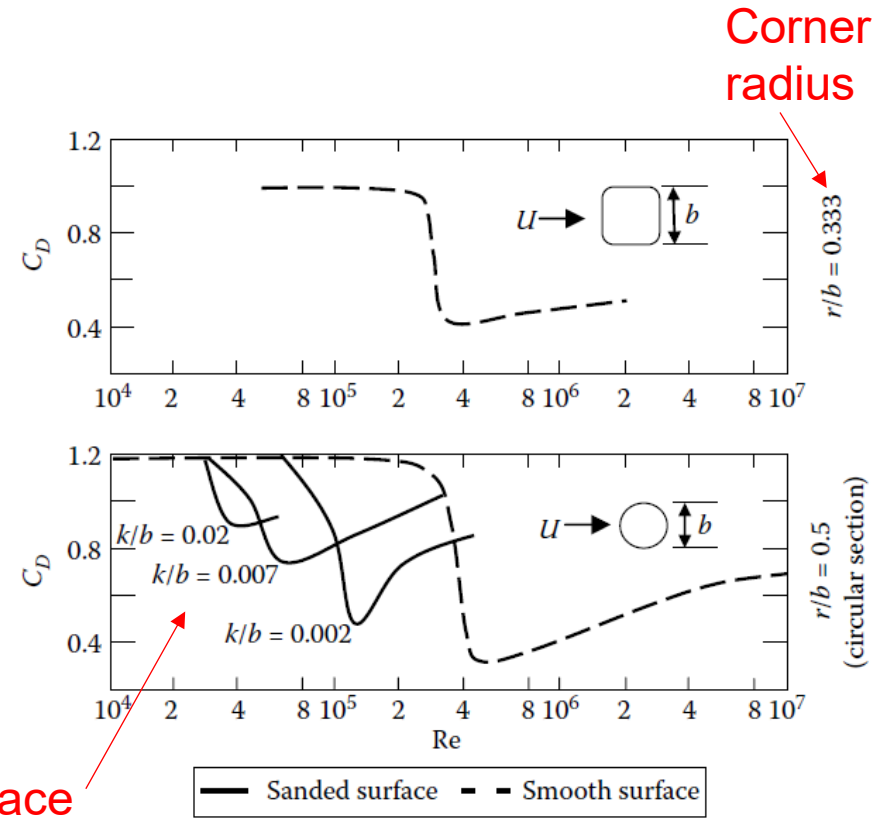
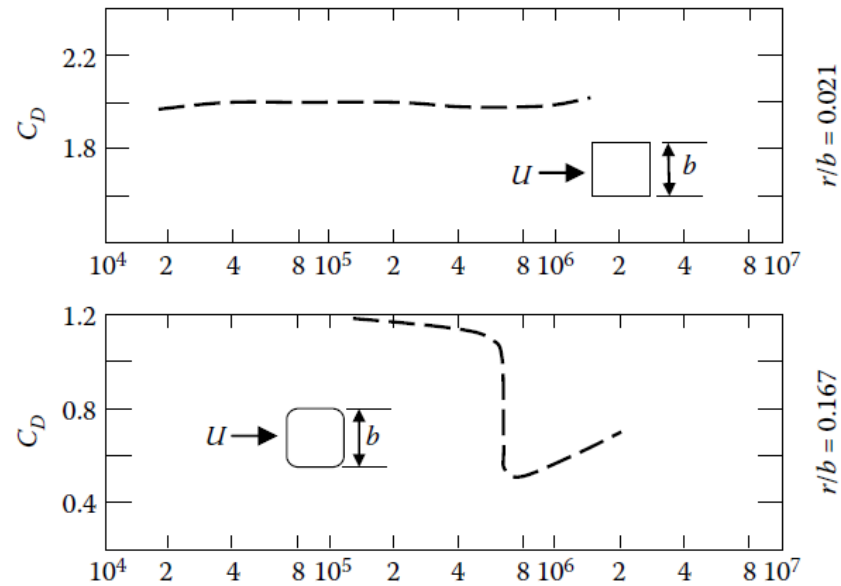


Streamline body



Bluff body

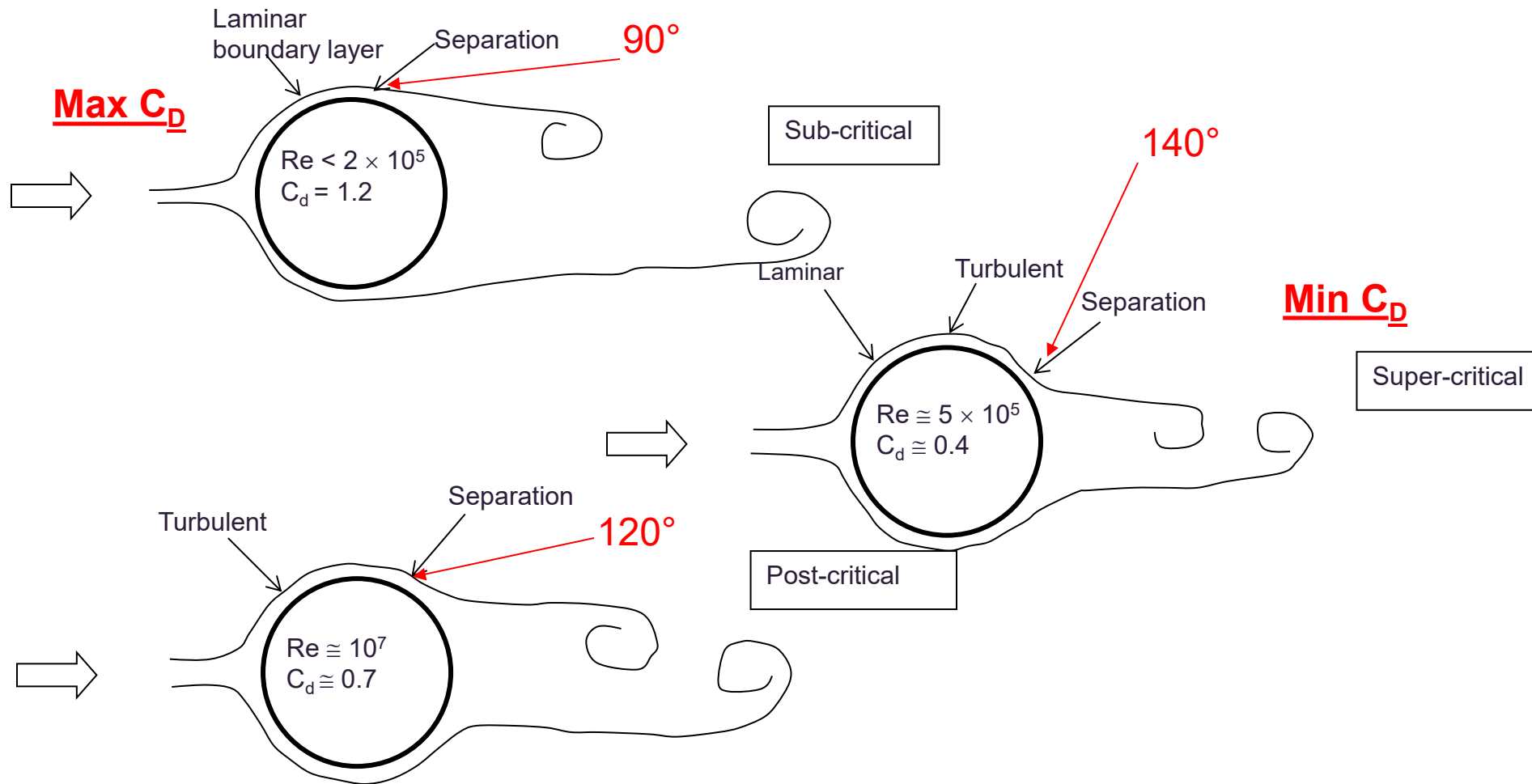
Drag examples (2D bodies)



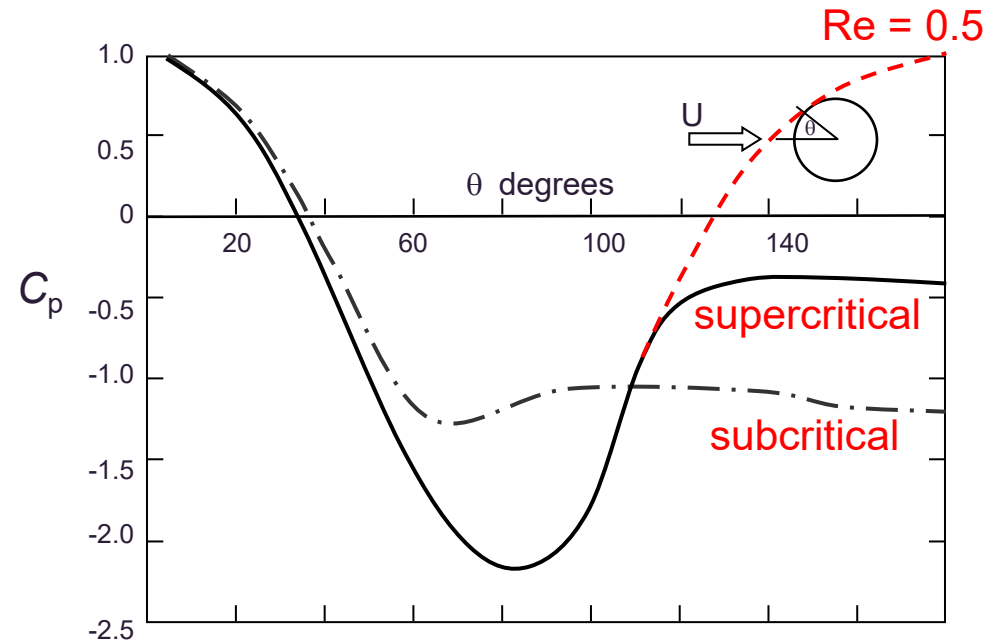
Surface roughness

Drag examples (2D bodies)

Circular cylinders (smooth flow)



Drag examples (2D bodies)



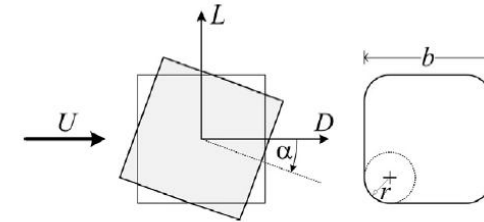
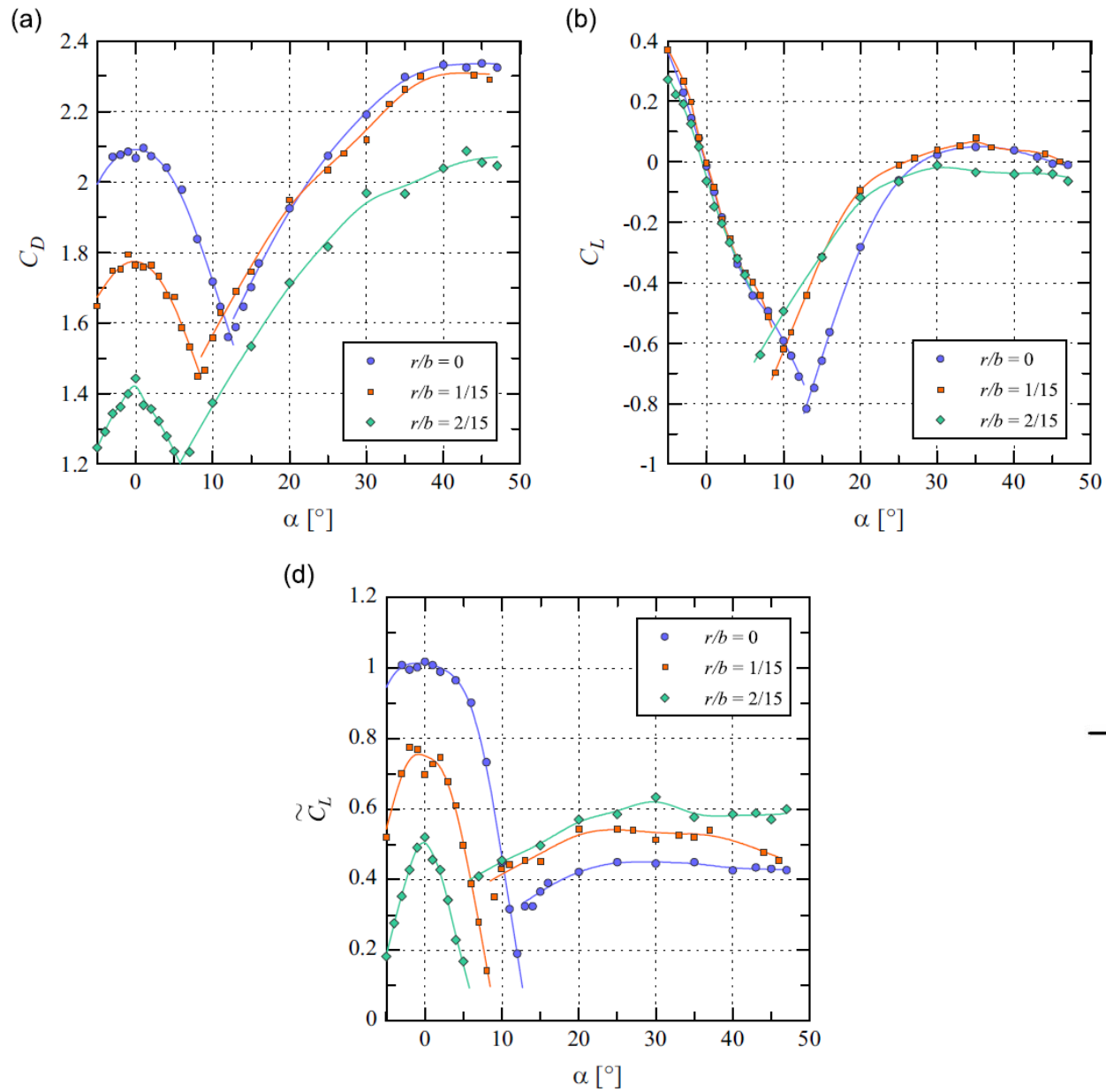
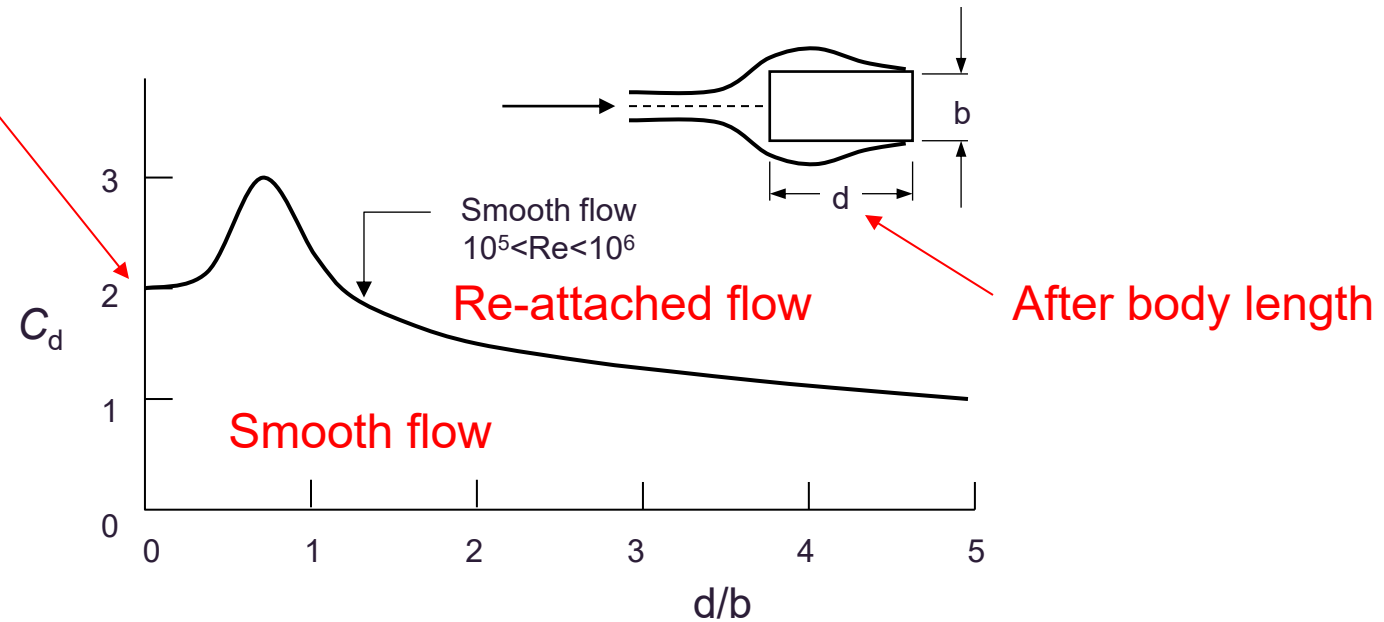


Fig. 1. Experimental setup.

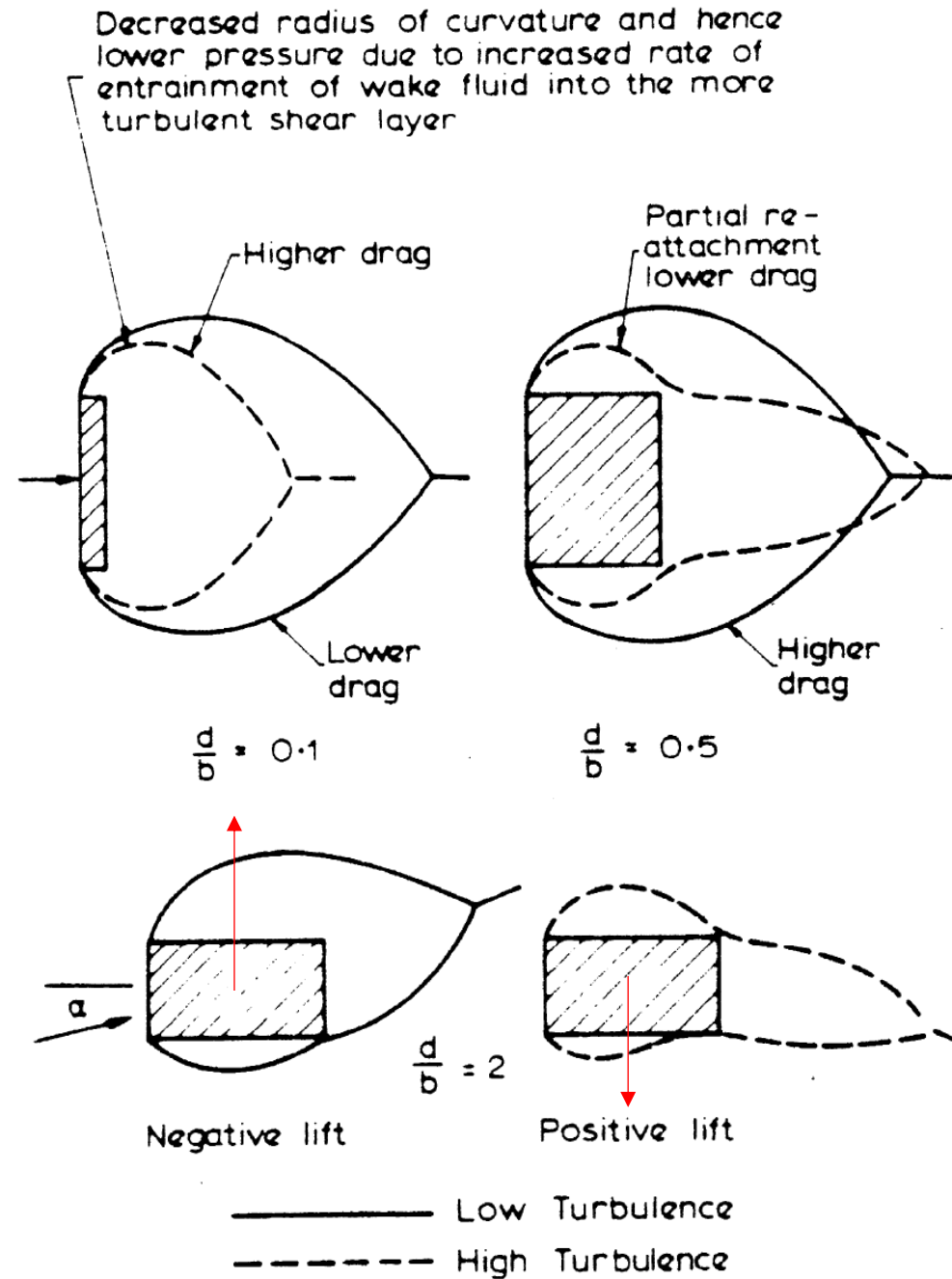
Fig. 2. Steady drag (a) and lift (b) coefficients, Strouhal number (c) and fluctuating lift coefficient (d) in smooth flow as functions of α . $Re = 3.7 \times 10^4$ for $r = 0$; $Re = 2.7 \times 10^4$ for $r/b = 1/15$ and $r/b = 2/15$.

Drag examples (2D bodies)

Plate ($C_D = 1.9$)



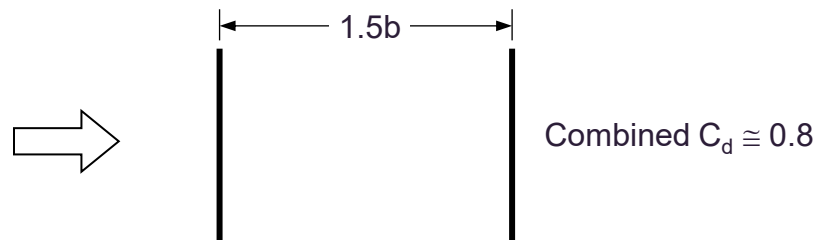
Maximum C_d at $d/b \approx 0.7$



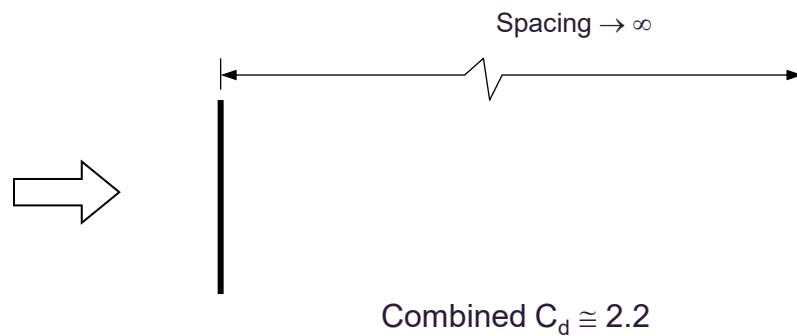
Drag examples (flat plates)



acts like a single plate

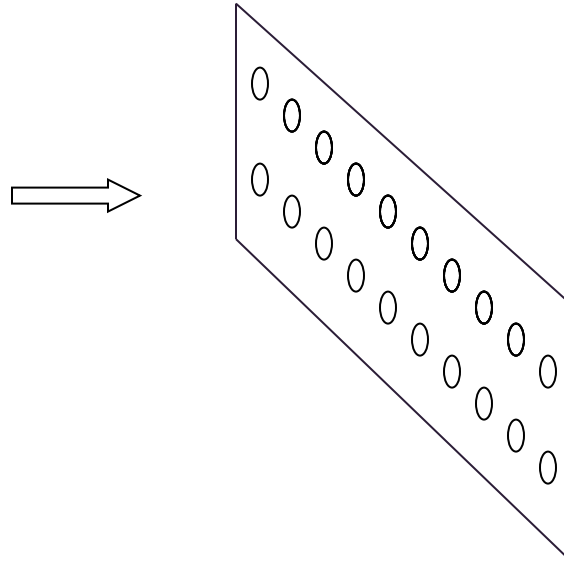


combined drag is *less than* single plate



acts like two single plates

Drag examples (flat plates)



$$C_{D,\delta} = C_D \cdot K_p$$

K_p = porosity factor,

$$K_p \cong 1 - (1 - \delta)^2$$

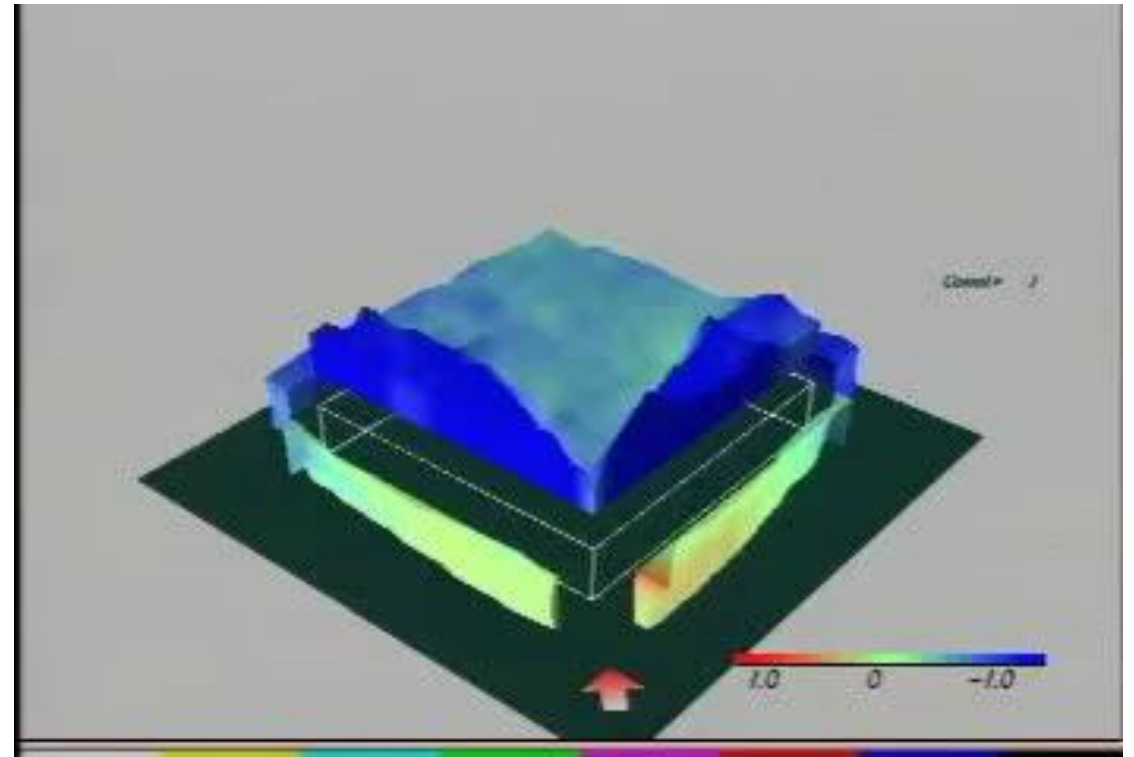
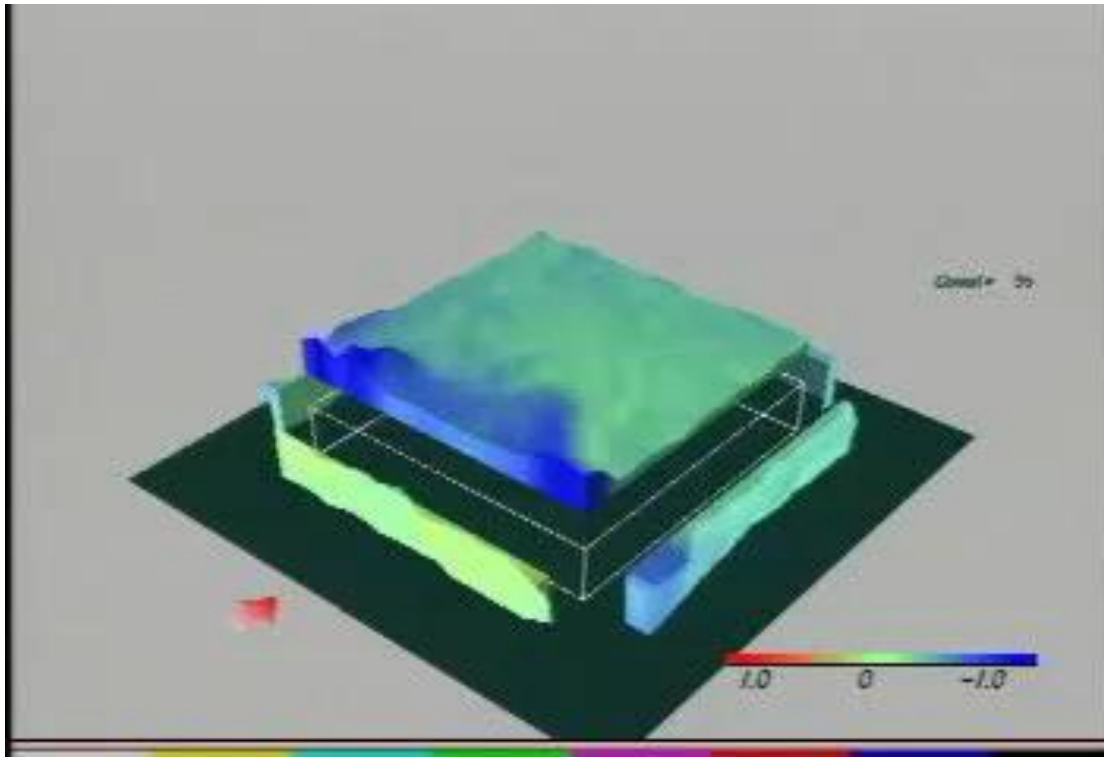
δ = solidity = solid area/total area

K_p is not sensitive to the shape of the opening (e.g. truss plates).

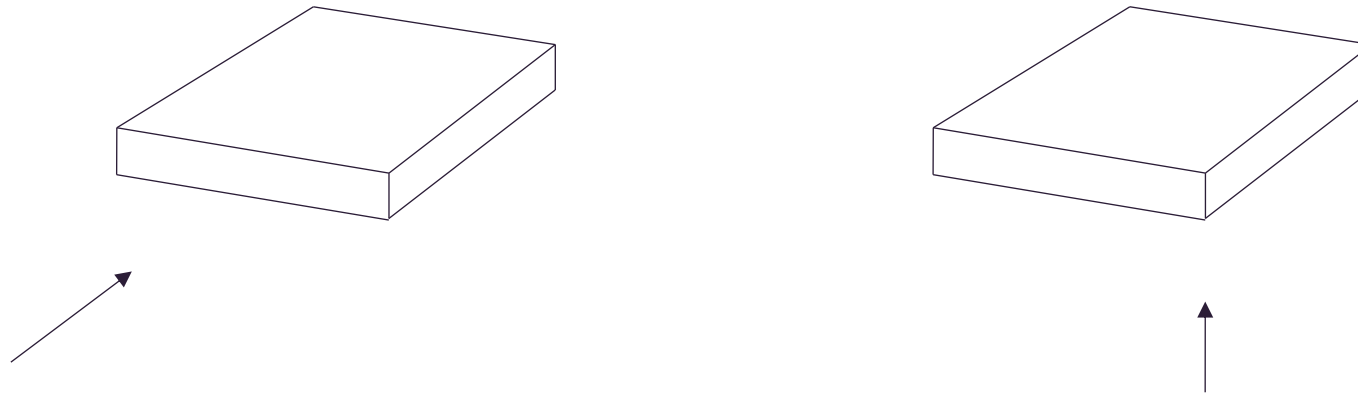
Fluctuating loads

- Free stream turbulence (buffeting)
 - Fluctuations in the approach flow transferred to the structure (quasi-steady assumption).
- Unsteadiness in bluff body forces
 - Driven by the fluctuating shear layer.
 - e.g. vortex shedding, re-attachment points.
- Aeroelastic forces
 - Forces due to the movement of the structure itself.
 - Only an issue for very flexible structures.
 - Very complicated, generally requires experimentation.

Fluctuating loads



Fluctuating loads



Have a go at drawing these 3D streamlines

Buffeting

Assumes pressures respond instantaneously to wind speed:

$$p(t) = \frac{1}{2}\rho U^2(t) \bar{C}_p(\theta + \alpha, \beta)$$

Inst. vertical dirn.
Inst. deviation from mean horizontal dirn.
Mean horizontal dirn.

dividing by $\frac{1}{2}\rho \bar{U}^2$

$$C_p(t, \theta + \alpha, \beta) = (U^2(t)/\bar{U}^2) \bar{C}_p(\theta + \alpha, \beta)$$

Then at time of peak gust (*influence of α and β assumed small*),

$$\widehat{C}_p(\theta) = (\widehat{U}^2(t)/\bar{U}^2) \bar{C}_p(\theta)$$

or

$$\hat{p} = \frac{1}{2}\rho \widehat{U}^2(t) \bar{C}_p(\theta)$$

Used in AS/NZS1170.2

Works well for the windward face, but not so well for those within the separated regions (body induced)

Body induced fluctuations

Vortex shedding

- Produced by the roll up of shear layers alternately on either side of a body.
- Occurs in laminar and turbulent flow
- If shedding frequency aligns with that of the body “lock-in” occurs
- As each vortex sheds a cross-wind load is applied to the structure

Vortex shedding generally quantified through the *Strouhal Number*, St

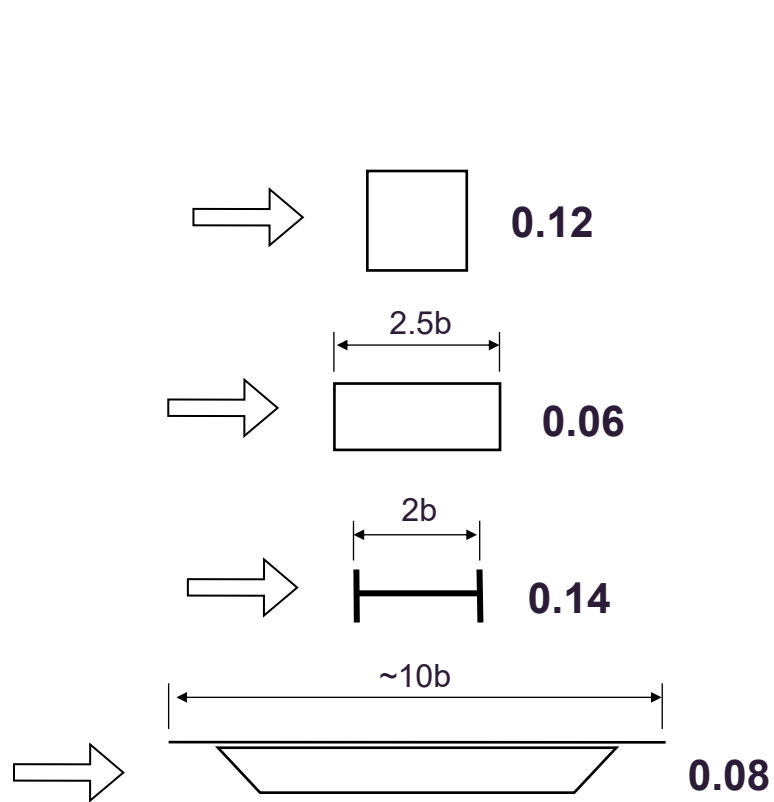
$$St = \frac{nL}{\overline{U}}$$

Frequency of shedding

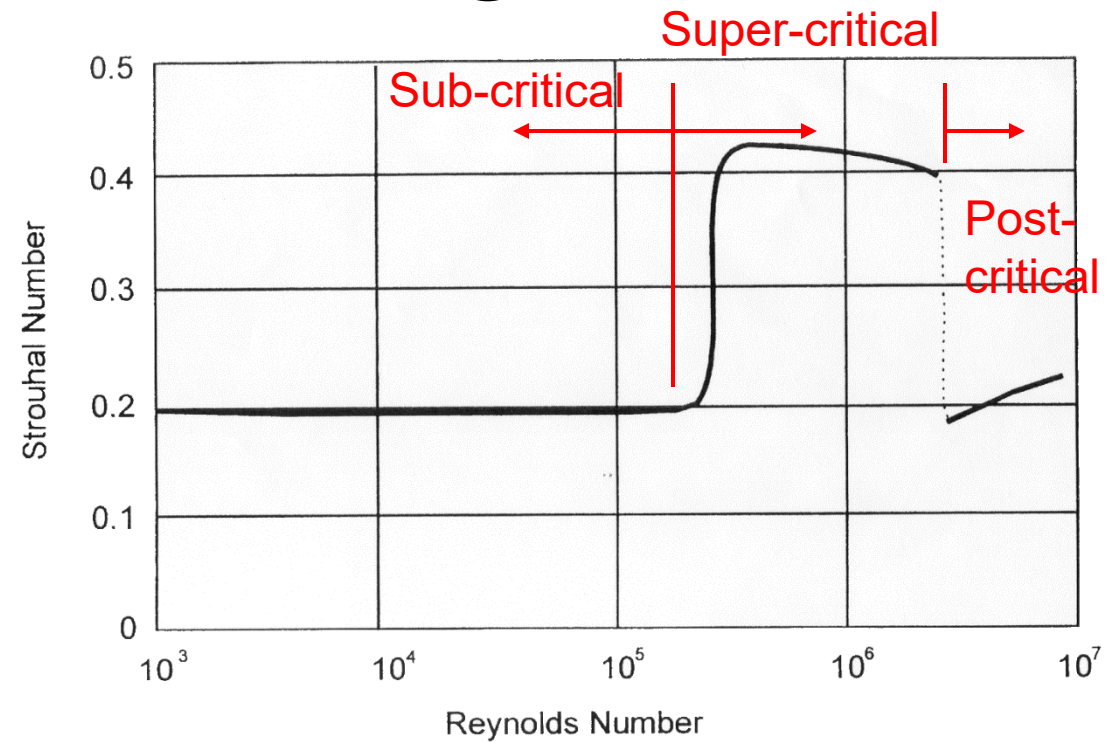
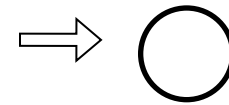
Length

Mean velocity

St examples



Independent of Re



Re dependent

St examples

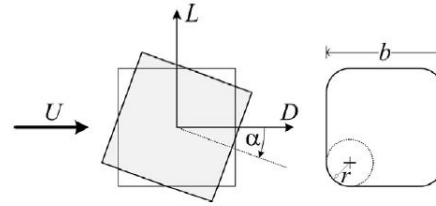
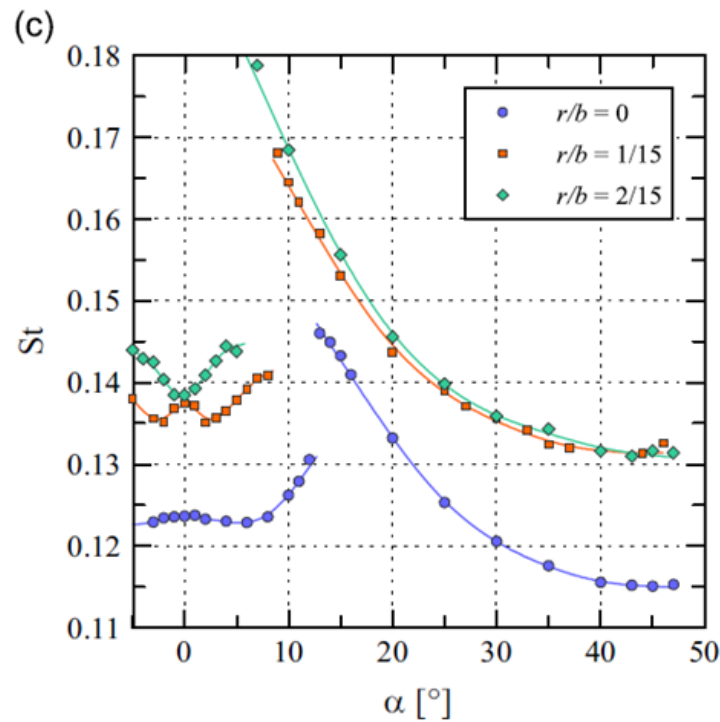
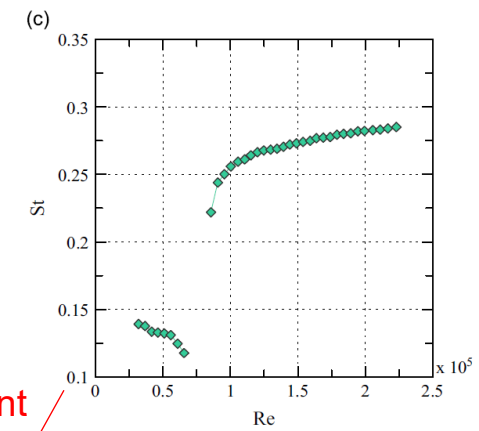
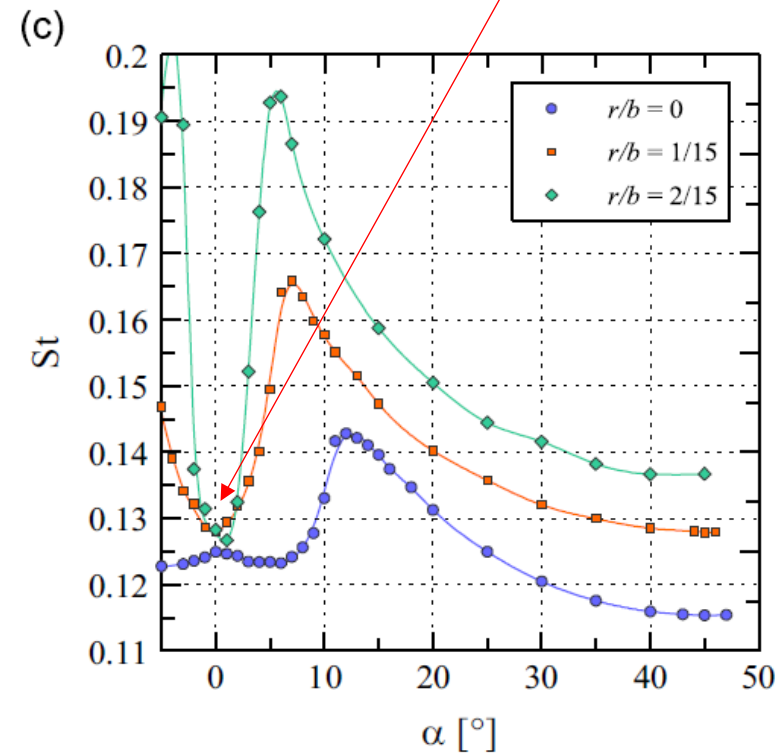


Fig. 1. Experimental setup.



Non-turbulent flow



Turbulent flow

Factors influencing C_p, F_D, F_L

$$C_p = f(\pi_1, \pi_2, \dots, \pi_n)$$

Pi terms

- L/L_{ref} – body dimensions (includes aspect ratio)
- α - Angle of wind attack
- h/z_0 – Jensen Number (h is height of structure)
- σ/U – turbulence intensities
- $\ell_u/L, \ell_v/L, \ell_w/L$ - turbulence length scale ratios
- $UL\rho/\mu$ – Reynolds number
- nL/U – Strouhal number (n is oscillation frequency)
- $U/(gL)^{1/2}$ – Froude number
- U/c – Mach Number (c is the speed of sound)

AS/NZS1170.2 [Sect. 5]

A modified quasi-steady approach, where K factors correct quasi-steady assumptions.

$$\hat{p} = \bar{C}_p (1/2) \rho_a \hat{U}^2 K_a K_l K_p K_c$$

Diagram illustrating the components of the modified quasi-steady approach equation:

- $\hat{p}$: Peak point pressure
- $\bar{C}_p$: Mean pressure coefficient
- $(1/2)$: Design wind speed
- ρ_a : Air density
- $\hat{U}^2$: Design wind speed
- K_a : Area reduction (not 100% correlated)
- K_l : Local correction (e.g. separated regions)
- K_p : Porosity
- K_c : Combination (loading due to more than 1 face)

Summary

- Overview of bluff body aerodynamics.
- Pressure coefficients and drag/lift forces.
- Fluctuating loads on bluff bodies
 - Buffeting
 - Body induced

Contact

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